

TASK 1 REPORT

CyberDefenders HawkEye Lab – Incident Analysis Report

Report	Task 1 - CyberDefenders HawkEye Lab Incident Analysis
Evidence Examined	stealer.pcap
Malware Family	HawkEye Keylogger Reborn v9
Tools Used	Wireshark, CyberChef, PowerShell

1. Executive Summary

A network forensic investigation was conducted on the packet capture file stealer.pcap to identify malicious activity within the organization's network. Analysis revealed that a workstation belonging to user roman.mcguire was infected with HawkEye Keylogger Reborn v9.

The malware collected sensitive credentials from browsers and email applications and exfiltrated the stolen information through an attacker-controlled SMTP server. The investigation successfully identified the victim system, attacker infrastructure, malware variant, and compromised credentials.

2. Investigation Details

Evidence Examined

stealer.pcap

Tools Used

- Wireshark
- CyberChef
- PowerShell

Methodology

- Analyzed network traffic using Wireshark.
- Examined Protocol Hierarchy and Endpoint statistics.
- Investigated SMTP communications.
- Reconstructed TCP streams.
- Decoded Base64-encoded content.
- Identified malware behavior and recovered stolen credentials.
- Documented Indicators of Compromise (IOCs).

3. Victim Information

Item	Value
Hostname	BEIJING-5CD1-PC
Internal IP Address	10.4.10.132
Public IP Address	173.66.146.112
User Account	roman.mcguire

4. Malware Information

Item	Value
Malware Family	HawkEye Keylogger Reborn v9
Malicious File	tkraw_Protected99.exe
MD5 Hash	71826ba081e303866ce2a2534491a2f7

Malware Capabilities

- Credential theft

- Browser password extraction
- Email credential theft
- Keylogging
- SMTP-based data exfiltration

5. SMTP Exfiltration Details

SMTP Server Information

Item	Value
Server IP	23.229.162.69
Hostname	p3plcpnl0413.prod.phx3.secureserver.net

Attacker Credentials

Item	Value
Username	sales.del@macwinlogistics.in
Password	Sales@23

Email Subject Used for Exfiltration

HawkEye Keylogger - Reborn v9 - Passwords Logs - roman.mcguire BEIJING-5CD1-PC - 173.66.146.112

6. Recovered Credentials

AOL Account

Item	Value
Username	roman.mcguire914@aol.com
Password	P@ssw0rd\$

Bank of America Account

Item	Value
Username	roman.mcguire
Password	P@ssw0rd\$

Microsoft Outlook Account

Item	Value
Email Address	roman.mcguire@pizzajukebox.com
Username	roman.mcguire
Password	P@ssw0rd\$
POP3 Server	pop.pizzajukebox.com
POP3 Port	995
SMTP Server	smtp.pizzajukebox.com
SMTP Port	587

7. Indicators of Compromise (IOCs)

Host Indicators

- Hostname: BEIJING-5CD1-PC
- Internal IP: 10.4.10.132
- Public IP: 173.66.146.112

Malware Indicators

- File Name: tkraw_Protected99.exe
- MD5 Hash: 71826ba081e303866ce2a2534491a2f7

Network Indicators

- SMTP Server IP: 23.229.162.69
- SMTP Hostname: p3plcpnl0413.prod.phx3.secureserver.net
- Exfiltration Email Account: sales.del@macwinlogistics.in

8. Attack Timeline

Stage 1 - Malware Download

The victim downloaded the malicious executable file tkraw_Protected99.exe.

Stage 2 - Malware Execution

The malware executed on the victim workstation.

Stage 3 - Credential Theft

Credentials from AOL, Bank of America, and Microsoft Outlook were extracted.

Stage 4 - Data Collection

The malware packaged the stolen information into a report.

Stage 5 - SMTP Exfiltration

The malware authenticated to the SMTP server and transmitted the stolen information to the attacker.

9. Findings

- The victim workstation was infected with HawkEye Keylogger Reborn v9.
- Sensitive credentials were successfully harvested.
- SMTP was used as the exfiltration channel.
- Attacker-controlled credentials were embedded within the malware configuration.
- Banking and email credentials were compromised.
- The attack reached the exfiltration stage successfully.

10. Recommendations

- Reset all compromised passwords immediately.
- Enable Multi-Factor Authentication (MFA).
- Reimage affected systems.

- Block malicious IP addresses and domains.
- Monitor outbound SMTP traffic.
- Deploy Endpoint Detection and Response (EDR) solutions.
- Conduct user awareness training.
- Perform regular threat hunting and log analysis.

11. Conclusion

The investigation confirmed that the victim workstation was compromised by HawkEye Keylogger Reborn v9. The malware successfully harvested sensitive credentials and exfiltrated them using attacker-controlled SMTP infrastructure. Through network forensic analysis, the victim system, malware variant, attacker infrastructure, and compromised accounts were successfully identified and documented.